Lecture plan · for the lecturer

# Squeeze and sample

University, vision 3

**Time**
45 min

**Materials**
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

## Learning goals

- Squeeze a picture into a short code and rebuild it.
- Say what a VAE adds: a cloud to sample from.
- Read the difference after one nudge.

## Activity: Leave it, then move it

1. Warm-up (5 min): Write “the latent space understands the picture” on the board, and ask who agrees.
2. Lab (15 min): Pairs leave the code and read the difference (0.025: the two bright specks are gone), then move one slider and read it again.
3. Explain (10 min): Encoder, decoder and reconstruction loss, and what a VAE’s cloud adds.
4. Repair (10 min): Strike the board sentence and write what the two numbers stand for.
5. Exit (5 min): Each student answers: does a nudged code rebuild or invent?

## Discussion

- What did the code throw away?

## Exit ticket

- A nudged code rebuilds, or invents?

**Try it in your next lecture**
Bring one forbidden sentence to your next lecture and have students repair it.

Seminar sheet · for students

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Name _____ Group _____ Date _____

**1** **The two-number code of the blob is…**

◯ A proof the picture is a photo ◯ The name of the object ◯ Where the brightness sits

**2** **Nudging the code gives a new picture; it does not repair the original pixels.**

◯ True ◯ False

**3** **Match the model.**

| | | |
|---|---|---|
| Autoencoder | • • • • | A short summary |
| Code | • • • • | A new code, decoded |
| Sample | • • • • | Squeeze, then rebuild |

**4** **Order one encode and decode.**

☐ Store those two numbers

☐ Paint a blob at that code

☐ Find where the brightness sits

Answers · for the lecturer

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## 1 Where the brightness sits

Encoding takes the centre of the brightness.

## 2 True

It paints a new blob at the new code. The exact pixels were never stored.

## 3 Autoencoder — Squeeze, then rebuild
## Code — A short summary
## Sample — A new code, decoded

A variational autoencoder (VAE) keeps a cloud of codes, so a new sample can be drawn from it.

## 4 Top to bottom: 2, 3, 1

The picture you get back is the decode, not the file you started with.

## For you to think about

**Which phrase will you ban on the slides?**